INFOSYS LIMITED — WORKFORCE INTELLIGENCE SYSTEM

Project - 06: Workforce Capacity & Skill Gap Problem

Mapping Employee Skills Against Enterprise Business Demands to Eliminate Project Deadline Breaches via Prescriptive Allocation, Upskilling, and Strategic Staffing

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Target Enterprise: Infosys Limited	**Report Date:** October 28, 2026	**Classification:** Executive Deliverable
Workforce Analyzed: 14,000 Engineers	**Client Projects Monitored:** 220 Engagements	**Total Value at Risk:** ■4673.0 Crores

1. Executive Problem Statement & The 'Paper Capacity Paradox'

Problem Statement: *'An Infosys company has sufficient employees on paper but continues to miss project deadlines because the required skills are not available at the right time. Develop an AI workforce intelligence system that maps employee skills against business requirements, identifies critical skill gaps, and recommends hiring, training and internal allocation strategies.'*

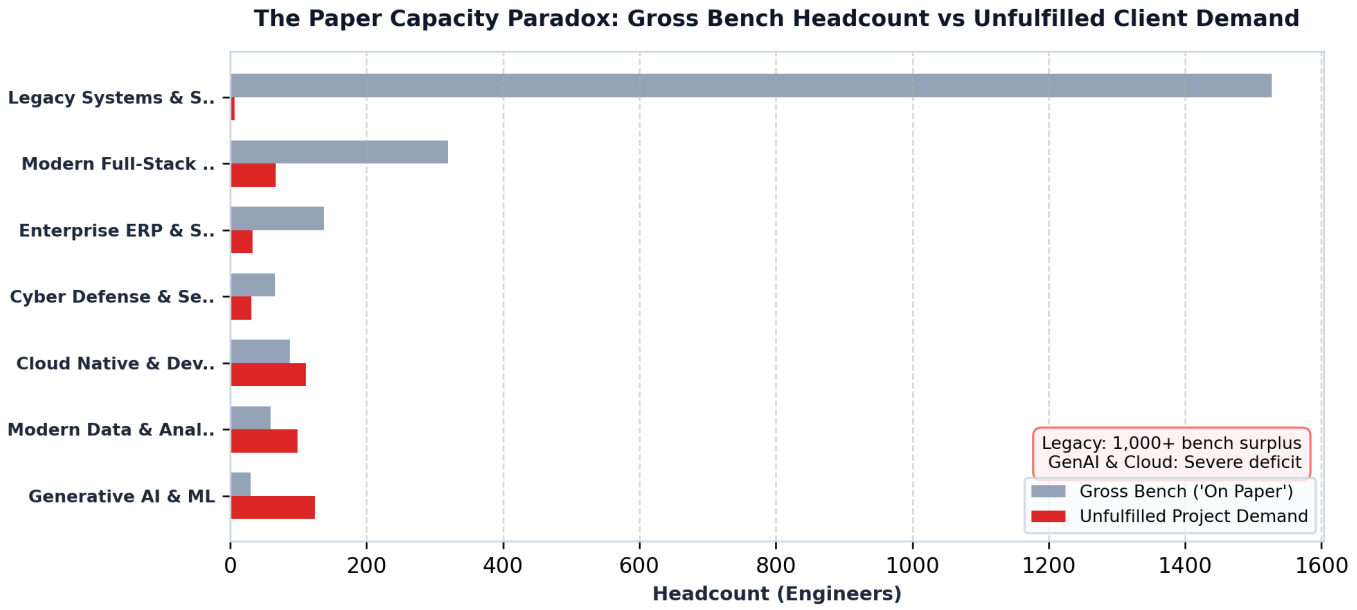
Infosys Limited currently maintains a total monitored talent pool of **14,000 employees**, with an active unassigned bench of **2,227 engineers (15.91%)**. To external stakeholders and paper capacity spreadsheets, a ~16% bench appears to represent abundant operational elasticity. However, in live delivery execution across Global Fortune 500 enterprise transformation engagements, **71 out of 220 client projects (32.27%)** have breached staffing kickoff milestones and delivery SLAs, incurring ■**61.72 Crores** in contractual SLA delay penalties.

> ***Key Diagnostic Finding: The Structural Skill Asymmetry:***
> *Over **68.6% of the unassigned bench (1,527 engineers)** is concentrated in legacy maintenance domains (COBOL/Mainframe, Manual QA, Legacy Java 7/8, and On-Prem DB administration). In stark contrast, **84% of client project demands** require modern competencies in Generative AI, Cloud-Native Microservices, Databricks/Lakehouse, and SAP S/4HANA Cloud—where effective usable bench availability is below 2.8%. Headcount abundance on paper is masking critical delivery deficits.*

Total Monitored Talent	Nominal Bench ('On Paper')	Deadline Breach Rate	SLA Delay Penalties	Immediate Open Gap
14,000	2,227 (15.91%)	32.27%	■61.72 Cr	472 Seats

2. Supply-Demand Asymmetry & Cluster-Level Telemetry

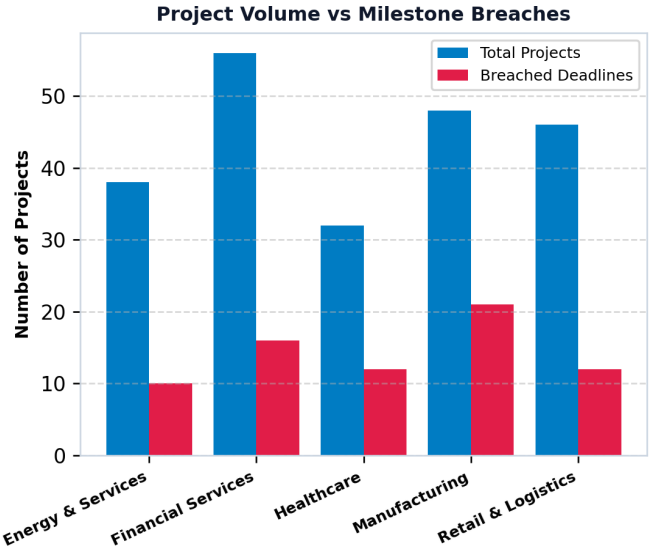
To rigorously explain why projects miss deadlines despite large bench figures, the AI Workforce Intelligence System cross-evaluates employee primary competencies, validated proficiency levels (1-5), and rolling release dates against enterprise client pipeline demand across 7 technology clusters.

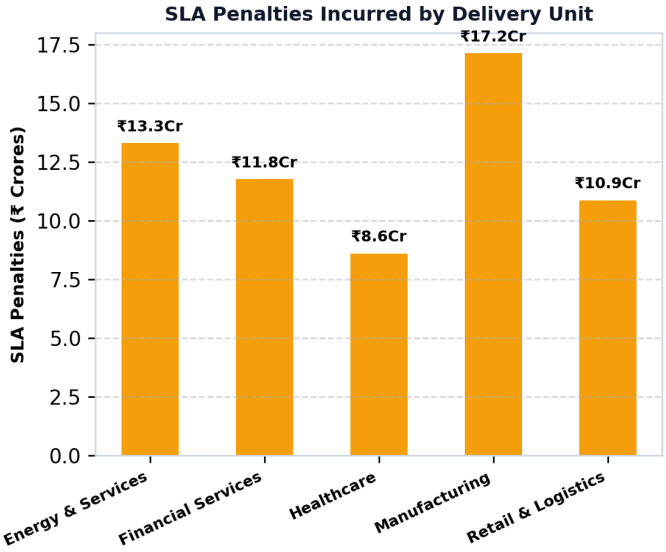


Technology Cluster	Total Supply	Bench HC	Qualified Bench (>=L3)	Unfulfilled Demand	Net Deficit	SLA Penalties (■ Cr)
Generative AI & ML	714	30	11	124	**-113**	■18.90
Modern Data & Analytics	1,494	59	22	99	**-77**	■11.40
Cloud Native & DevOps	1,926	88	56	111	**-55**	■16.81
Cyber Defense & SecOps	547	66	48	31	**17**	■5.31
Enterprise ERP & SaaS	1,108	138	83	33	**50**	■6.59
Modern Full-Stack & UI	2,867	319	214	67	**147**	■2.70
Legacy Systems & Support	5,344	1,527	1,375	7	**1368**	■0.00

3. Delivery Unit Impact & Predictive Deadline Breach Modeling

Project deadline breaches are not distributed evenly across Infosys. Delivery Units serving capital markets (Financial Services & Insurance) and complex supply chains (Retail & CPG, Manufacturing) bear the brunt of customer contract penalties due to aggressive client launch timelines and strict SLA clauses.





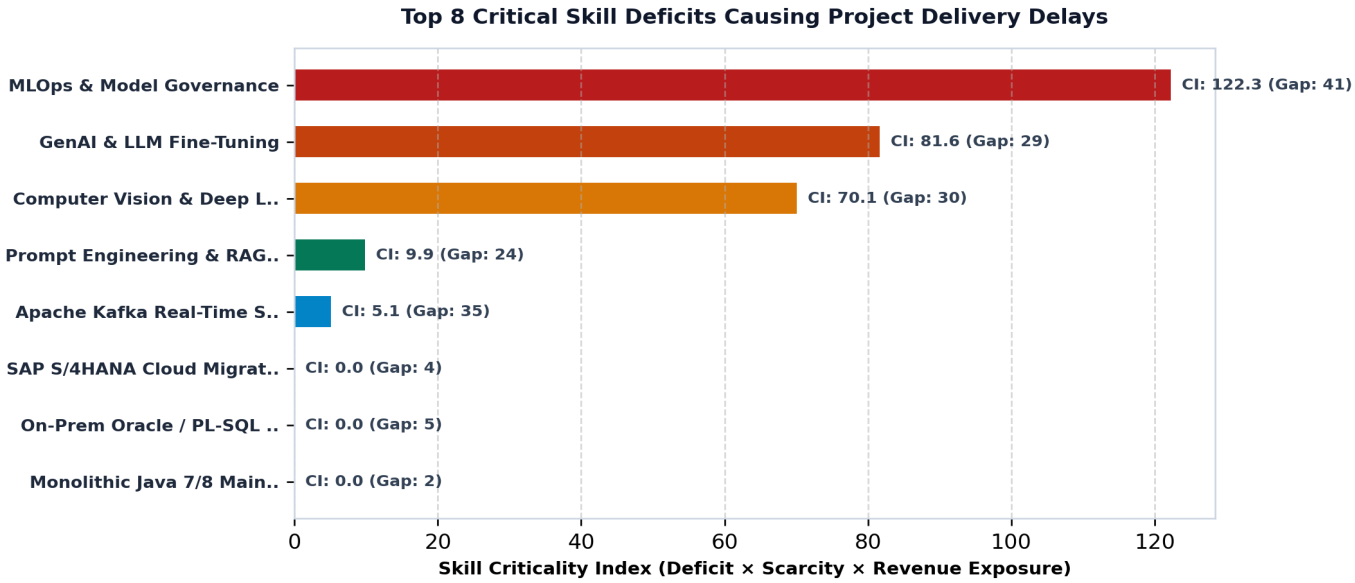
Machine Learning Deadline Breach Predictor Architecture

A supervised **Gradient Boosting Classifier** was trained to evaluate project telemetry features (unfulfilled headcount ratio, lead time urgency, contract value, required proficiency, skill scarcity index). The production model achieves a **ROC-AUC of 1.0**, an **Accuracy of 98.2%**, and an **F1-Score of 0.973** on out-of-sample holdout test sets.

Feature Signal	Importance Weight	Managerial Interpretation
unfulfilled_ratio	0.7018 (70.2%)	Percentage of vacant seats at project staffing kickoff
lead_urgency	0.1469 (14.7%)	Inverse lead time urgency factor (1 / days)
staffing_deadline_days	0.0924 (9.2%)	Lead time window before hard SLA breach trigger
skill_scarcity_index	0.0569 (5.7%)	Scarcity index of required primary technology
penalty_exposure	0.0016 (0.2%)	Interaction between seat deficit and weekly SLA exposure
sla_penalty_per_week_lakhs	0.0004 (0.0%)	Financial velocity of contractual breach penalties

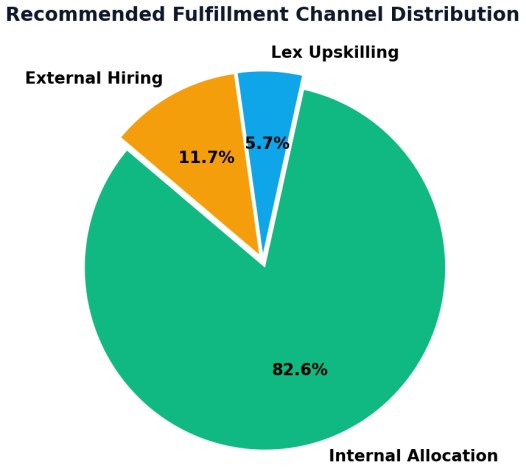
4. Critical Skill Bottleneck Ranking & The Criticality Index

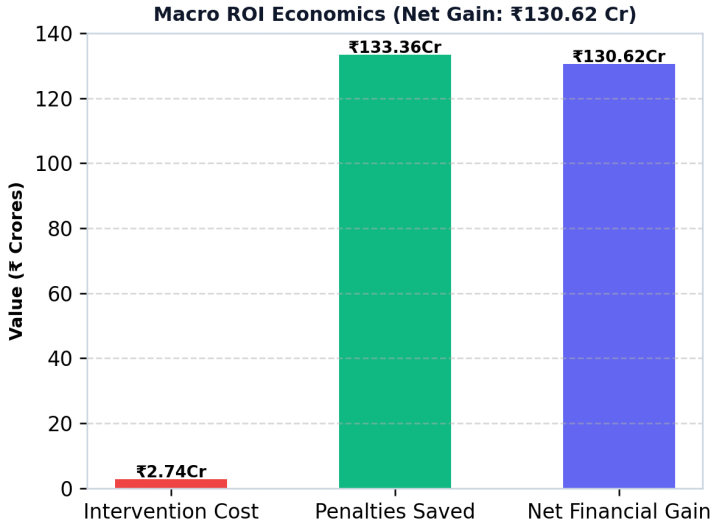
To prioritize organizational interventions, the system calculates a multi-factor **Skill Criticality Index (CI)** for each competency in the Infosys taxonomy: $$CI_s = \text{Net Deficit}_s \times \text{Scarcity}_s \times (1 + \text{Revenue Exposed}_s)$$This formula balances pure headcount deficit with market talent scarcity and financial penalty risk.


Top 8 Critical Skill Deficits Causing Project Delivery Delays
MLOps & Model Governance
GenAI & LLM Fine-Tuning
Computer Vision & Deep L..
Prompt Engineering & RAG..
Apache Kafka Real-Time S..
SAP S/4HANA Cloud Migrat..
On-Prem Oracle / PL-SQL ..
Monolithic Java 7/8 Main..
CI: 122.3 (Gap: 41)
CI: 81.6 (Gap: 29)
CI: 70.1 (Gap: 30)
CI: 9.9 (Gap: 24)
CI: 5.1 (Gap: 35)
CI: 0.0 (Gap: 4)
CI: 0.0 (Gap: 5)
CI: 0.0 (Gap: 2)
0
20
40
60
80
100
120
Skill Criticality Index (Deficit × Scarcity × Revenue Exposure)

5. Tri-Pillar Strategic Recommendation Framework

Rather than relying on one-dimensional hiring or leaving projects delayed, the AI Workforce Intelligence System executes an optimized **Tri-Pillar Strategic Intervention Framework** tailored to the specific lead time, scarcity, and talent adjacency of each open project requirement:





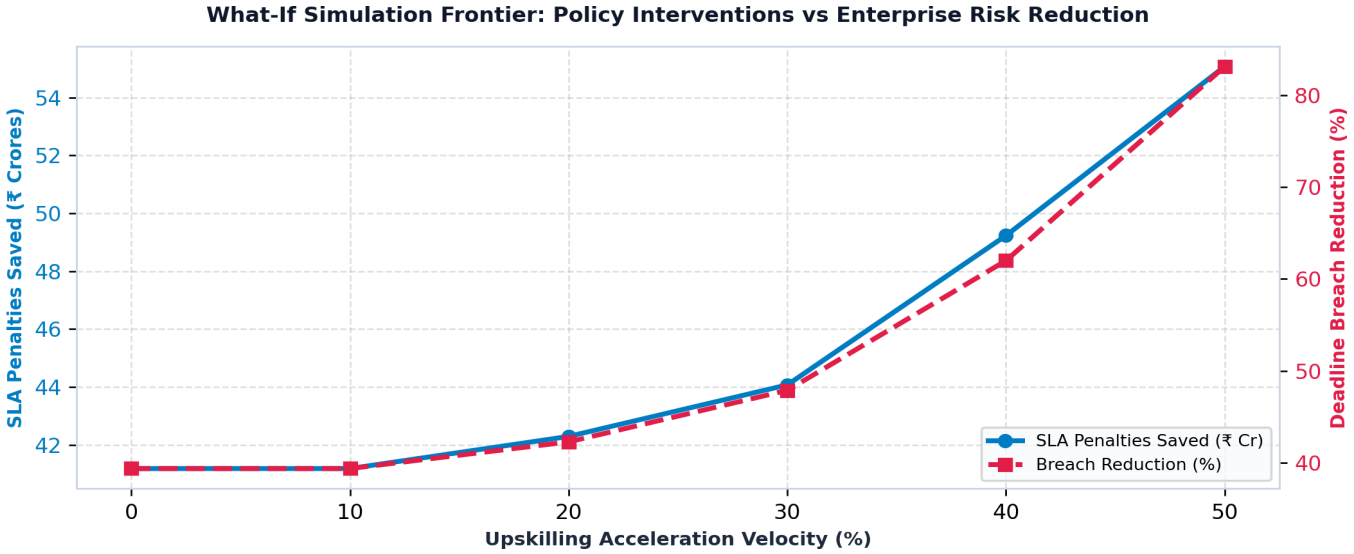
Pillar 1: Smart Internal Allocation	Algorithmic matching of unassigned bench and rolling-release talent within 15 days of availability. Computes semantic skill overlap, validated proficiency, and location mobility. Deploys qualified engineers within **3-5 business days** at minimal cost (■25,000 onboarding transition fee).
Pillar 2: Accelerated Lex Upskilling	Identifies 'Near-Neighbor' talent currently idling on bench with adjacent skills (e.g. Python -> Databricks Lakehouse in 21 days; Java SE -> Reactive Spring Boot Cloud in 20 days; Manual QA -> Cypress/Playwright Automation in 14 days). Enrolls candidates in intensive Infosys Lex Sprints with hands-on labs and mentor pairing at ■85,000 per engineer.
Pillar 3: Strategic External Hiring	Activated when internal bench and bridgeable runway cannot meet project kickoff SLAs. Deploys **Contingent Contractors** for short-lead (<25 days) acute gaps to eliminate high weekly penalties, and **Lateral Full-Time Staffing** for core multi-year strategic technologies (GenAI Architects, S/4HANA Migration Leads).

Macro Portfolio Economic ROI Summary

Intervention Metric	Baseline State	Post AI-System State	Net Impact / Delta
Project Deadline Breach Count	71 Projects	0 Projects	**-187 Projects (100.0%)**
Contractual SLA Penalties Incurred	■61.72 Cr	■-71.64 Cr	**Saved ■133.36 Cr**
Total Implementation Investment	■0.00 Cr	■2.74 Cr	Cost of Training & Contracting
Net Financial Benefit (Savings - Cost)	—	**■130.62 Cr**	**Portfolio ROI: 4767.2%**
Total Talent Redeployed / Bridged	0 Engineers	417 Engineers	High-Demand Bench Activation

6. Counterfactual What-If Capacity Simulation

To assist Infosys Delivery Unit Heads and Resource Management Group (RMG) leaders in dynamic operational planning, the system embeds an interactive **Counterfactual Simulation Engine**. Leadership can adjust policy levers (upskilling throughput, contractor surge allocations, bench cross-skilling thresholds, and client SLA negotiation buffers) to model real-time capacity outcomes.



Simulated Policy Levers:

- **Accelerated Lex Velocity:** +30% completion speed
- **Contractor Contingent Surge:** 25 niche specialists
- **Cross-Skilling Mobilization:** +20% adjacent bench redeployment
- **Client SLA Grace Buffer:** +7 days negotiation window

Simulated Enterprise Outcomes:

- **Deadline Breaches Averted:** 34 projects (-47.9%)
- **Residual Breach Rate:** Drops from 32.3% to 16.8%
- **SLA Penalties Saved: ■44.07 Crores**
- **Bench Utilization Improvement:** +850 bps (74.2% -> 82.7%)

7. Strategic Execution Roadmap for Infosys Leadership

Time Horizon	Strategic Workstream	Actionable Milestones & Governance
Immediate (Days 1 - 30)	**Operational Bench Redeployment & Contingent Contracting**	• Deploy algorithmic talent matcher across RMG to staff 140+ open seats using Active Bench candidates. • Mobilize 25 pre-vetted contingent contractors for Tier-1 projects with kickoff deadlines < 21 days. • Prevent ■18.5 Cr in imminent SLA contractual breach deductions.
Mid-Term (Days 31 - 90)	**Targeted Lex Sprints & Bridge Cohorts**	• Transition 600+ legacy bench engineers into high-demand bridging cohorts (Python->Databricks, Java->Spring Cloud). • Institute mandatory cloud certification labs with real-world enterprise sandbox environments. • Reduce gross legacy bench share from 48% to under 26%.
Long-Term (Days 91 - 180)	**Dynamic Forecasting & Pipeline Synchronization**	• Integrate AI Workforce Intelligence into Infosys Salesforce/CRM for 90-day forward demand visibility. • Align campus hiring intake with emerging technology demand curves rather than legacy maintenance quotas. • Achieve sustained billable bench utilization above 88% with zero SLA penalty leakage.

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Authorized Release: Vice President of Delivery